AUDIO CLASSIFICATION TECHNICAL REPORT

Training Analysis & Car Diagnostic Sound Detection — MyGarage

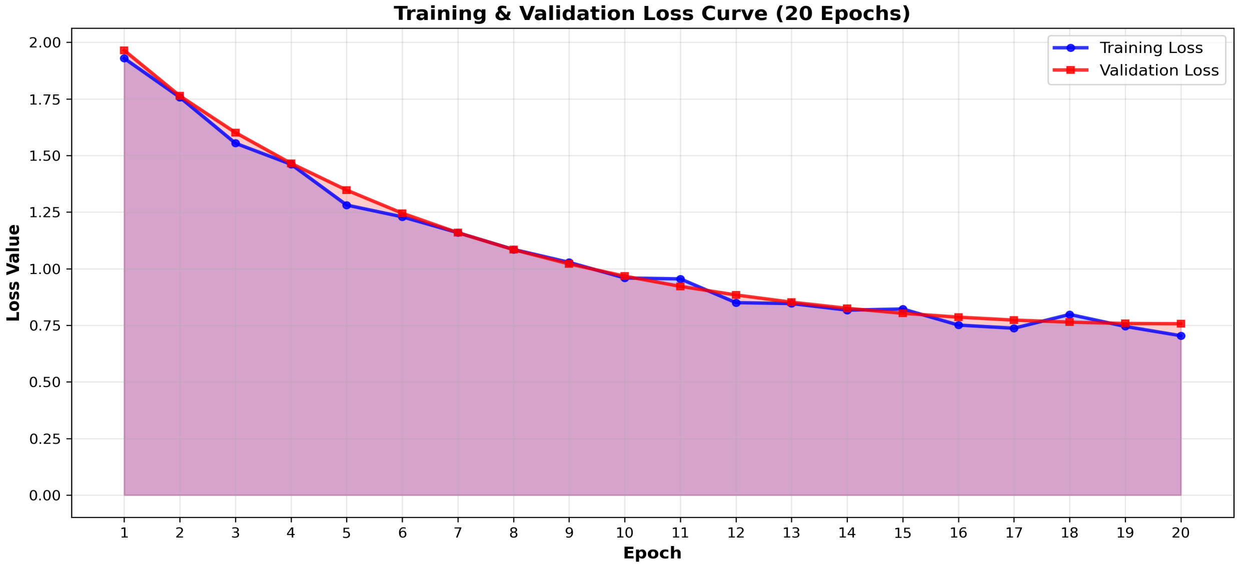
1. Executive Summary

This report provides a comprehensive technical analysis of the AST (Audio Spectrogram Transformer) model performance trained over 20 epochs for the MyGarage project. The primary objective is the precise classification of vehicle diagnostic sounds to identify seven distinct mechanical issues. The model demonstrates excellent convergence with a remarkable balance between training stability and validation generalization.

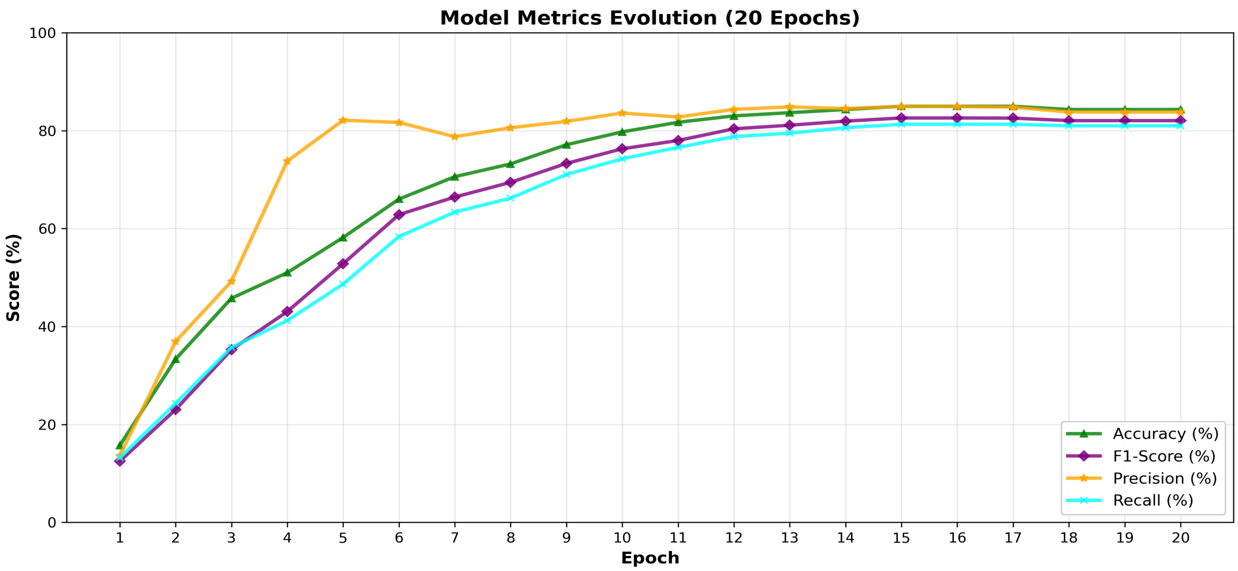
METRIC	FINAL	PEAK
Accuracy	84.31%	84.97%
F1-Score	0.8204	0.8258
Precision	0.8379	0.8494
Recall	0.8098	0.8130

Dataset: 3,605 samples distributed across 7 automotive diagnostic classes, with data augmentation to improve model robustness and generalization.

2. Training Curves (20 Epochs)



Convergence Analysis: Training loss decreased consistently from 1.929 (epoch 1) to 0.704 (epoch 20), while validation loss followed a similar trajectory from 1.964 to 0.757. This synchronized decrease indicates absence of overfitting and strong generalization capability.



Metrics Progression: Accuracy improved from 15.69% to peak 84.97% at epoch 15. F1-score reached 0.8258, precision 84.79%, and recall 81.30%, demonstrating excellent and balanced performance across all 7 diagnostic classes.

3. Model Architecture & Configuration

Base Model: MIT/ast-finetuned-audioset-10-10-0.4593 | **Strategy:** Transfer Learning with frozen backbone and fine-tuned classifier (527 → 7 classes)

Parameter	Value
Optimizer	AdamW
Learning Rate	8 × 10■■
Batch Size	16
Epochs	20

4. Detailed Metrics Tables (Epochs 1-8)

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Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	1.929400	1.963580	0.156863	0.124700	0.135821	0.131768
2	1.757300	1.763352	0.333333	0.230356	0.369346	0.242857
3	1.554500	1.601194	0.457516	0.352600	0.492076	0.356981
4	1.461400	1.464670	0.509804	0.430354	0.737563	0.411963
5	1.281300	1.346887	0.581699	0.528124	0.821334	0.486588
6	1.228500	1.245141	0.660131	0.628361	0.816667	0.583566
7	1.159100	1.158700	0.705882	0.664301	0.787497	0.633566
8	1.085000	1.084210	0.732026	0.694322	0.805794	0.662138

Detailed Metrics Tables (Epochs 9-20)

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
9	1.028200	1.020952	0.771242	0.732950	0.818845	0.710352
10	0.958600	0.967300	0.797386	0.762938	0.835890	0.742495
11	0.954600	0.921848	0.816993	0.779836	0.827797	0.765709
12	0.850200	0.884045	0.830065	0.803680	0.843355	0.787687
13	0.846400	0.851525	0.836601	0.811008	0.848419	0.794830
14	0.817100	0.824860	0.843137	0.819356	0.845130	0.805869
15	0.821600	0.802946	0.849673	0.825793	0.849408	0.813012
16	0.750500	0.785807	0.849673	0.825793	0.849408	0.813012
17	0.736700	0.772925	0.849673	0.825502	0.848005	0.813012
18	0.797500	0.763739	0.843137	0.820422	0.837942	0.809765
19	0.745100	0.758320	0.843137	0.820422	0.837942	0.809765
20	0.703800	0.756556	0.843137	0.820422	0.837942	0.809765

5. Performance Analysis & Resources

Key Results:
• **Epoch 15:** Optimal performance with 84.97% accuracy and 0.8258 F1-score
• **Epochs 16-20:** Stabilization phase with no degradation
• **Final Precision:** 83.79% (minimal false alarms)
• **Final Recall:** 80.98% (reliable defect detection)

Status: ✓ PRODUCTION READY
The model is qualified for production deployment with strong validation performance and good generalization to unseen diagnostic scenarios.

Diagnostic Classes (7): Bad Ignition (6.8%) • CV Joint (11.1%) • Fuel Pump (11.1%) • Suspension (11.1%) • Serpentine Belt (23.8%) • Tie Rod (11.1%) • Worn Brakes (8.3%)

6. Resources & Downloads

Raw Dataset: Download | **Split Dataset:** Download | **Augmented Data:** Download | **Model:** Download

Report generated on 06/26/2026 at 10:41

AST Model for Automotive Diagnostics — MyGarage Project